

Optimization Lecture 9

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Outline

Terminology and assumptions

Generic descent method

Gradient Descent Method

Steepest descent method

Newton's method

Self-concordance functions

Implementation

What we learned during the last two lectures

- ▶ Lagrangian and dual functions
- ▶ dual problem
- ▶ KKT conditions
- ▶ sensitivity analysis
- ▶ Problem reformulations
- ▶ Theorems of alternatives

Terminology and assumptions

Unconstrained minimization

- ▶ unconstrained minimization problem

$$\text{minimize } f(x)$$

- ▶ we assume
 - ▶ f convex, twice continuously differentiable (hence $\text{dom } f$ open)
 - ▶ optimal value $p^* = \inf_x f(x)$ is attained at x^* (not necessarily unique)
- ▶ optimality condition is $\nabla f(x) = 0$
- ▶ minimizing f is the same as solving $\nabla f(x) = 0$ (a set of n equations with n unknowns)

Quadratic functions

- ▶ convex quadratic: $f(x) = (1/2)x^T P x + q^T x + r, P \succeq 0$
- ▶ we can solve exactly via linear equations

$$\nabla f(x) = P x + q = 0$$

- ▶ very important since a function can be approximated by quadratic locally.
- ▶ Optimal solution=? $p^* = ?$

Why quadratic function is important?

Iterative methods

- ▶ for most non-quadratic functions, we use iterative methods
- ▶ these produce a sequence of points $x^{(k)} \in \text{dom } f, k = 0, 1, \dots$
- ▶ $x^{(0)}$ is the initial point or **starting point**
- ▶ $x^{(k)}$ is the k th iterate
- ▶ we hope that the method converges, i.e.,

$$f\left(x^{(k)}\right) \rightarrow p^*, \quad \nabla f\left(x^{(k)}\right) \rightarrow 0$$

Initial point and sublevel set

- ▶ iterative algorithms require a starting point $x^{(0)}$ such that
 - ▶ $x^{(0)} \in \text{dom } f$
 - ▶ sublevel set $S = \{x \mid f(x) \leq f(x^{(0)})\}$ is closed
- ▶ 2nd condition is
 - ▶ equivalent to condition that $\text{epi } f$ is closed
 - ▶ true if $\text{dom } f = \mathbf{R}^n$
 - ▶ true if $f(x) \rightarrow \infty$ as $x \rightarrow \text{bd dom } f$
- ▶ examples of differentiable functions with closed sublevel sets:

$$f(x) = \log \left(\sum_{i=1}^m \exp(a_i^T x + b_i) \right), \quad f(x) = - \sum_{i=1}^m \log(b_i - a_i^T x)$$

Strong convexity and implications

- ▶ f is **strongly convex** on S if there exists an $m > 0$ such that

$$\nabla^2 f(x) \geq mI \text{ for all } x \in S$$

(related to curvature)

- ▶ same as $f(x) - (m/2)\|x\|_2^2$ is convex
- ▶ if f is strongly convex, for $x, y \in S$,

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) + \frac{m}{2}\|x - y\|_2^2$$

- ▶ hence, S is bounded
- ▶ we conclude $p^* > -\infty$, and for $x \in S$,

$$f(x) - p^* \leq \frac{1}{2m}\|\nabla f(x)\|_2^2$$

(Pf: set y =the optimal solution to minimize the RHS...)

- ▶ useful as stopping criterion (if you know m , which usually you do not)

Generic descent method

Descent methods

- ▶ descent methods generate iterates as

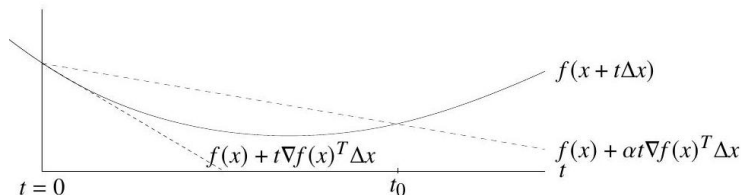
$$x^{(k+1)} = x^{(k)} + t^{(k)} \Delta x^{(k)}$$

with $f(x^{(k+1)}) < f(x^{(k)})$ (hence the name)

- ▶ other notations: $x^+ = x + t\Delta x$, $x := x + t\Delta x$
- ▶ $\Delta x^{(k)}$ is the step, or search direction
- ▶ $t^{(k)} > 0$ is the step size, or step length
- ▶ from convexity, $f(x^+) < f(x)$ implies $\nabla f(x)^T \Delta x < 0$ (why?)
- ▶ Δx is a descent direction.

Line search types

- ▶ exact line search: $t = \operatorname{argmin}_{t \geq 0} f(x + t\Delta x)$
- ▶ backtracking line search (with parameters $\alpha \in (0, 1/2), \beta \in (0, 1)$ why?)
- ▶ starting at $t = 1$, repeat $t := \beta t$ until $f(x + t\Delta x) < f(x) + \alpha t \nabla f(x)^T \Delta x$
- ▶ graphical interpretation: reduce t (i.e., backtrack) until $t \leq t_0$



The obtained t satisfies $\beta t_0 \leq t \leq 1$ (why?)

Generic decent method

Generic descent method

given a starting point $x \in \text{dom } f$.

repeat

1. Determine a descent direction Δx .
2. Line search. Choose a step size $t > 0$.
3. Update. $x := x + t\Delta x$.

until stopping criterion is satisfied.

Gradient Descent Method

Gradient descent method

- ▶ general descent method with $\Delta x = -\nabla f(x)$

given a starting point $x \in \text{dom } f$.

repeat

1. $\Delta x := -\nabla f(x)$.
2. Line search. Choose step size t via exact or backtracking line search.
3. Update. $x := x + t\Delta x$.

until stopping criterion is satisfied.

- ▶ stopping criterion usually of the form $\|\nabla f(x)\|_2 \leq \epsilon$
- ▶ convergence result: for strongly convex f ,

$$f(x^{(k)}) - p^* \leq c^k (f(x^{(0)}) - p^*)$$

$c \in (0, 1)$ depends on $m, x^{(0)}$, line search type (how to prove it?)

- ▶ very simple, but can be very slow

Outline of Pf (assume f is strongly convex and the exact search is used):

- ▶ Let $g(t) = f(x - t\nabla f(x))$.
- ▶ In S , There is $M > 0$ s.t. $\nabla^2 f(x) \leq M$. Thus:

$$g(t) \leq f(x) - t\|\nabla f(x)\|_2^2 + \frac{Mt^2}{2}\|\nabla f(x)\|_2^2$$

(Taylor extension)

- ▶ do minimization on RHS: $t = 1/M$:

$$f(x^+) = g(t_{\text{exact}}) \leq f(x) - \frac{1}{2M}\|\nabla f(x)\|_2^2$$

- ▶ Noting $f(x) - p^* \leq \frac{1}{2m}\|\nabla f(x)\|_2^2$, we have

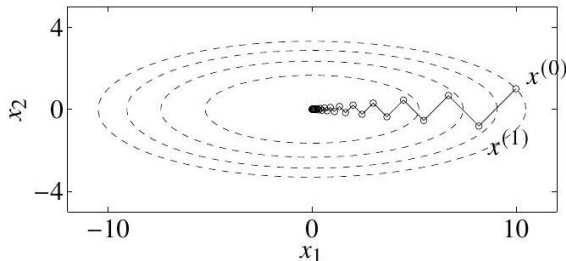
$$f(x^+) - p^* \leq (1 - m/M)(f(x) - p^*)$$

Example: Quadratic function on $\mathbf{R}^2$

- ▶ take $f(x) = (1/2)(x_1^2 + \gamma x_2^2)$, with $\gamma > 0$
- ▶ with exact line search, starting at $x^{(0)} = (\gamma, 1)$:

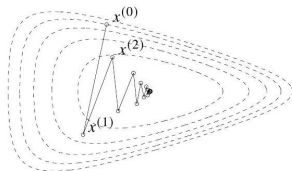
$$x_1^{(k)} = \gamma \left(\frac{\gamma - 1}{\gamma + 1} \right)^k, \quad x_2^{(k)} = \left(-\frac{\gamma - 1}{\gamma + 1} \right)^k$$

- ▶ very slow if $\gamma \gg 1$ or $\gamma \ll 1$ (condition number $\gg 1$)
- ▶ example for $\gamma = 10$ at right
- ▶ called zig-zagging

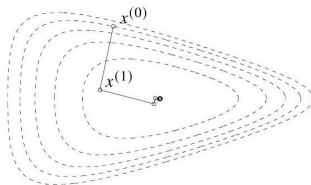


Example: Nonquadratic function on $\mathbf{R}^2$

$$f(x_1, x_2) = e^{x_1+3x_2-0.1} + e^{x_1-3x_2-0.1} + e^{-x_1-0.1}$$



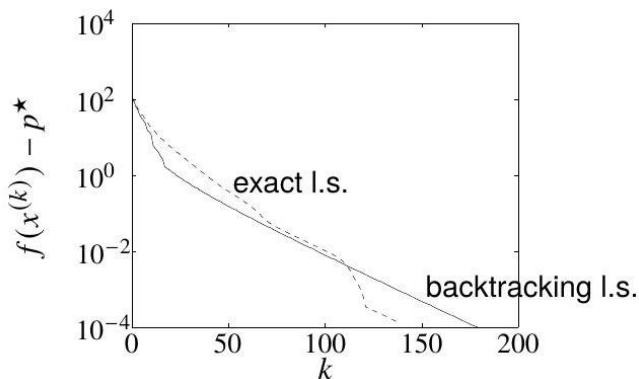
backtracking line search



exact line search

Example: A problem in $\mathbf{R}^{100}$

► $f(x) = c^T x - \sum_{i=1}^{500} \log(b_i - a_i^T x)$



- linear convergence, i.e., a straight line on a semilog plot
- exercise: do it using cvx.

Steepest descent method

Steepest descent method

- ▶ normalized steepest descent direction (at x , for norm $\|\cdot\|$):

$$\Delta x_{\text{nsd}} = \operatorname{argmin} \{ \nabla f(x)^T v \mid \|v\| = 1 \}$$

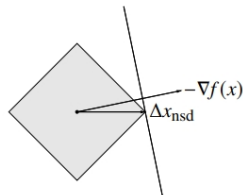
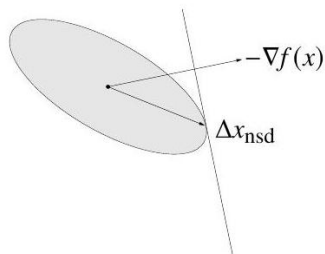
- ▶ interpretation: for small v , $f(x + v) \approx f(x) + \nabla f(x)^T v$;
- ▶ direction Δx_{nsd} is unit-norm step with most negative directional derivative
- ▶ (unnormalized) steepest descent direction: $\Delta x_{\text{sd}} = \|\nabla f(x)\|_* \Delta x_{\text{nsd}}$
- ▶ satisfies $\nabla f(x)^T \Delta x_{\text{sd}} = -\|\nabla f(x)\|_*^2$
- ▶ $\|a\|_* = \sup \{ a^T b \mid \|b\| = 1 \}$

steepest descent method

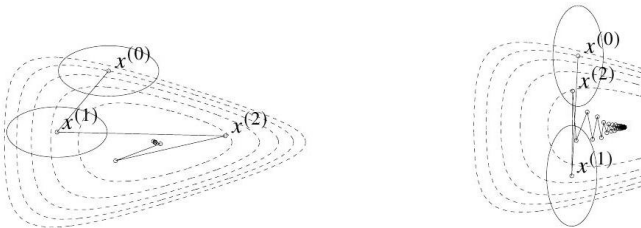
- ▶ general descent method with $\Delta x = \Delta x_{\text{sd}}$
- ▶ convergence properties similar to gradient descent

Examples

- ▶ Euclidean norm: $\Delta x_{\text{sd}} = -\nabla f(x)$
- ▶ quadratic norm
 $\|x\|_P = (x^T P x)^{1/2} \ (P \in \mathbf{S}_{++}^n) : \Delta x_{\text{sd}} = -P^{-1} \nabla f(x)$
- ▶ ℓ_1 -norm: $\Delta x_{\text{sd}} = -(\partial f(x)/\partial x_i) e_i$, where $|\partial f(x)/\partial x_i| = \|\nabla f(x)\|_\infty$
- ▶ unit balls, normalized steepest descent directions for quadratic norm and ℓ_1 -norm:



Choice of norm for steepest descent

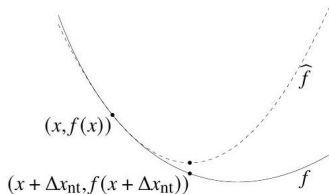


- ▶ steepest descent with backtracking line search for two quadratic norms
- ▶ ellipses show $\{x \mid \|x - x^{(k)}\|_P = 1\}$
- ▶ interpretation of steepest descent with quadratic norm $\|\cdot\|_P$: gradient descent after change of variables $\bar{x} = P^{1/2}x$
- ▶ shows choice of P has strong effect on speed of convergence
- ▶ Example Quadratic function on R^2 .

Newton's method

- ▶ Newton step is $\Delta x_{\text{nt}} = -\nabla^2 f(x)^{-1} \nabla f(x)$
- ▶ interpretation: $x + \Delta x_{\text{nt}}$ minimizes second order approximation

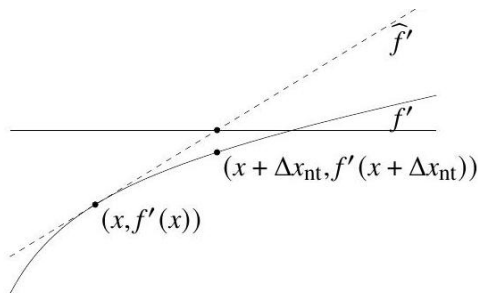
$$\hat{f}(x + v) = f(x) + \nabla f(x)^T v + \frac{1}{2} v^T \nabla^2 f(x) v$$



Another interpretation

- $x + \Delta x_{\text{nt}}$ solves linearized optimality condition

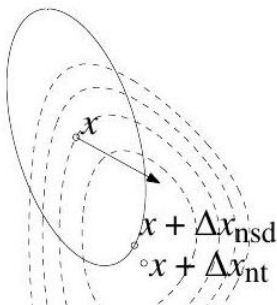
$$\nabla f(x + v) \approx \nabla \hat{f}(x + v) = \nabla f(x) + \nabla^2 f(x)v = 0$$



And one more interpretation

- ▶ Δx_{nt} is steepest descent direction at x in local Hessian norm

$$\|u\|_{\nabla^2 f(x)} = (u^T \nabla^2 f(x) u)^{1/2}$$



- ▶ dashed lines are contour lines of f ; ellipse is $\{x + v \mid v^T \nabla^2 f(x) v = 1\}$
- ▶ arrow shows $-\nabla f(x)$

Newton decrement

Newton decrement is $\lambda(x) = (\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x))^{1/2}$

- ▶ a measure of the proximity of x to x^*
- ▶ gives an estimate of $f(x) - p^*$, using quadratic approximation $\hat{f}$:

$$f(x) - \inf_y \hat{f}(y) = \frac{1}{2} \lambda(x)^2$$

- ▶ equal to the norm of the Newton step in the quadratic Hessian norm

$$\lambda(x) = (\Delta x_{\text{nt}}^T \nabla^2 f(x) \Delta x_{\text{nt}})^{1/2}$$

- ▶ directional derivative in the Newton direction:
 $\nabla f(x)^T \Delta x_{\text{nt}} = -\lambda(x)^2$

Newton's method

given a starting point $x \in \text{dom } f$, tolerance $\epsilon > 0$.

repeat

1. Compute the Newton step and decrement.

$$\Delta x_{\text{nt}} := -\nabla^2 f(x)^{-1} \nabla f(x); \quad \lambda^2 := \nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x)$$

2. Stopping criterion. quit if $\lambda^2/2 \leq \epsilon$.
3. Line search. Choose step size t by backtracking line search.
4. Update. $x := x + t\Delta x_{\text{nt}}$.

- affine invariant, i.e., independent of linear changes of coordinates
Newton iterates for $\tilde{f}(y) = f(Ty)$ with starting point $y^{(0)} = T^{-1}x^{(0)}$
are $y^{(k)} = T^{-1}x^{(k)}$

Classical convergence analysis of Newton's method

Assumptions

- ▶ f strongly convex on S with constant m
- ▶ $\nabla^2 f$ is Lipschitz continuous on S , with constant $L > 0$:

$$\|\nabla^2 f(x) - \nabla^2 f(y)\|_2 \leq L\|x - y\|_2$$

(L measures how well f can be approximated by a quadratic function)

Outline: there exist constants $\eta \in (0, m^2/L)$, $\gamma > 0$ such that

- ▶ if $\|\nabla f(x^k)\|_2 \geq \eta$, then $f(x^{(k+1)}) - f(x^{(k)}) \leq -\gamma$
- ▶ if $\|\nabla f(x^k)\|_2 < \eta$, then

$$\frac{L}{2m^2} \|\nabla f(x^{(k+1)})\|_2 \leq \left(\frac{L}{2m^2} \|\nabla f(x^{(k)})\|_2 \right)^2$$

Damped Newton phase ($\|\nabla f(x)\|_2 \geq \eta$)

- ▶ most iterations require backtracking steps
- ▶ function value decreases by at least γ
- ▶ if $p^* > -\infty$, this phase ends after at most $(f(x^{(0)}) - p^*) / \gamma$ iterations

Quadratically convergent phase ($\|\nabla f(x)\|_2 < \eta$)

- ▶ all iterations use step size $t = 1$
- ▶ $\|\nabla f(x)\|_2$ converges to zero quadratically: if $\|\nabla f(x^{(k)})\|_2 < \eta$, then

$$\frac{L}{2m^2} \|\nabla f(x^{(l)})\|_2 \leq \left(\frac{L}{2m^2} \|\nabla f(x^{(k)})\|_2 \right)^{2^{l-k}} \leq \left(\frac{1}{2} \right)^{2^{l-k}}, \quad l \geq k$$

Remember:

$$f(x) - p^* \leq \frac{1}{2m} \|\nabla f(x)\|_2^2$$

Then

$$f(x^{(l)}) - p^* \leq \frac{2m^3}{L^2} (1/2)^{2^{l-k+1}}$$

Roughly, the number of correct digits doubles at each generation (quadratic convergence)

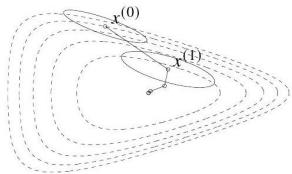
Conclusion: number of iterations until $f(x) - p^* \leq \epsilon$ is bounded above by

$$\frac{f(x^{(0)}) - p^*}{\gamma} + \log_2 \log_2 (\epsilon_0 / \epsilon)$$

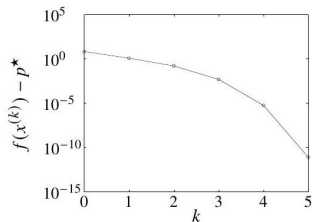
- ▶ γ, ϵ_0 are constants that depend on $m, L, x^{(0)}$
- ▶ second term is small (of the order of 6) and almost constant for practical purposes
- ▶ in practice, constants m, L (hence γ, ϵ_0) are usually unknown
- ▶ provides qualitative insight in convergence properties (i.e., explains two algorithm phases)

Example: $\mathbf{R}^2$

► $f(x) = c^T x - \sum_{i=1}^{500} \log(b_i - a_i^T x)$

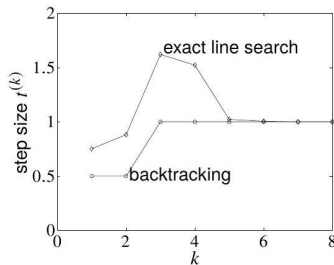
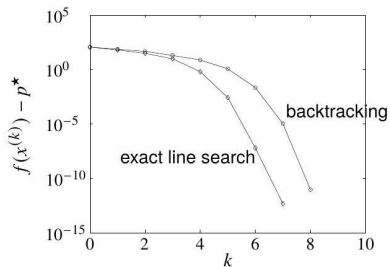


- backtracking parameters $\alpha = 0.1, \beta = 0.7$
- converges in only 5 steps
- quadratic local convergence



Example in $\mathbf{R}^{100}$

► $f(x) = c^T x - \sum_{i=1}^{500} \log(b_i - a_i^T x)$

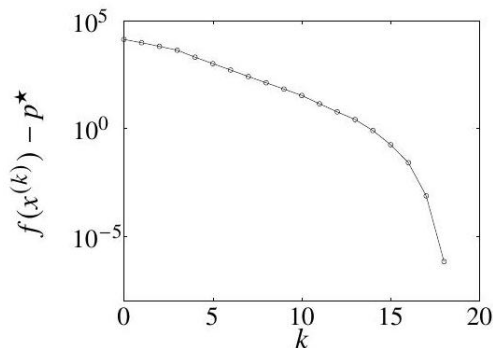


- backtracking parameters $\alpha = 0.01, \beta = 0.5$
- backtracking line search almost as fast as exact l.s. (and much simpler)
- clearly shows two phases in algorithm

Example in $\mathbf{R}^{10000}$

(with sparse a_i)

$$f(x) = - \sum_{i=1}^{10000} \log(1 - x_i^2) - \sum_{i=1}^{100000} \log(b_i - a_i^T x)$$



- ▶ backtracking parameters $\alpha = 0.01, \beta = 0.5$.
- ▶ performance similar as for small examples

Self-concordance functions

Why self-concordance

Shortcomings of classical convergence analysis

- ▶ depends on unknown constants ($m, L, \dots$)
- ▶ bound is not affinely invariant, although Newton's method is

convergence analysis via self-concordance (Nesterov and Nemirovski)

- ▶ does not depend on any unknown constants
- ▶ gives affine-invariant bound
- ▶ applies to special class of convex functions ('self-concordant' functions)
- ▶ developed to analyze polynomial-time interior-point methods for convex optimization

Def of Self-concordant functions

definition

- ▶ convex $f : \mathbf{R} \rightarrow \mathbf{R}$ is self-concordant if $|f'''(x)| \leq 2f''(x)^{3/2}$ for all $x \in \text{dom } f$
- ▶ $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is self-concordant if $g(t) = f(x + tv)$ is self-concordant for all $x \in \text{dom } f, v \in \mathbf{R}^n$

Examples on $\mathbf{R}$

- ▶ linear and quadratic functions
- ▶ negative logarithm $f(x) = -\log x$
- ▶ negative entropy plus negative logarithm: $f(x) = x \log x - \log x$

affine invariance: if $f : \mathbf{R} \rightarrow \mathbf{R}$ is s.c., then $\tilde{f}(y) = f(ay + b)$ is s.c.:

$$\tilde{f}'''(y) = a^3 f'''(ay + b), \quad \tilde{f}''(y) = a^2 f''(ay + b)$$

Self-concordant calculus

properties

- ▶ preserved under positive scaling $\alpha \geq 1$, and sum
- ▶ preserved under composition with affine function
- ▶ if g is convex with $\text{dom } g = \mathbf{R}_{++}$ and $|g'''(x)| \leq 3g''(x)/x$ then
$$f(x) = \log(-g(x)) - \log x$$

is self-concordant

examples: properties can be used to show that the following are s.c.

- ▶ $f(x) = -\sum_{i=1}^m \log(b_i - a_i^T x)$ on $\{x \mid a_i^T x < b_i, i = 1, \dots, m\}$
- ▶ $f(X) = -\log \det X$ on $\mathbf{S}_{++}^n$
- ▶ $f(x) = -\log(y^2 - x^T x)$ on $\{(x, y) \mid \|x\|_2 < y\}$

Convergence analysis for self-concordant functions

Summary: there exist constants $\eta \in (0, 1/4]$, $\gamma > 0$ such that

► if $\lambda(x) > \eta$, then

$$f(x^{(k+1)}) - f(x^{(k)}) \leq -\gamma$$

► if $\lambda(x) \leq \eta$, then

$$2\lambda(x^{(k+1)}) \leq \left(2\lambda(x^{(k)})\right)^2$$

(η and γ only depend on backtracking parameters α, β)

complexity bound: number of Newton iterations bounded by

$$\frac{f(x^{(0)}) - p^*}{\gamma} + \log_2 \log_2(1/\epsilon)$$

for $\alpha = 0.1, \beta = 0.8, \epsilon = 10^{-10}$, bound = $375 (f(x^{(0)}) - p^*) + 6$

Numerical example

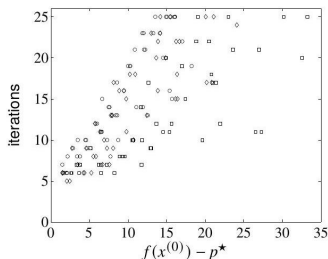
150 randomly generated instances of

$$\text{minimize } f(x) = - \sum_{i=1}^m \log(b_i - a_i^T x)$$

○ : $m = 100, n = 50$

□ : $m = 1000, n = 500$

◇ : $m = 1000, n = 50$



- ▶ number of iterations much smaller than $375 (f(x^{(0)}) - p^*) + 6$
- ▶ bound of the form $c (f(x^{(0)}) - p^*) + 6$ with smaller c (empirically) valid

Implementation

Implementation

main effort in each iteration: evaluate derivatives and solve Newton system

$$H\Delta x = -g$$

where $H = \nabla^2 f(x)$, $g = \nabla f(x)$
via Cholesky factorization

$$H = LL^T, \quad \Delta x_{\text{nt}} = -L^{-T}L^{-1}g, \quad \lambda(x) = \|L^{-1}g\|_2$$

- ▶ cost $(1/3)n^3$ flops for unstructured system
- ▶ cost $\ll (1/3)n^3$ if H sparse, banded